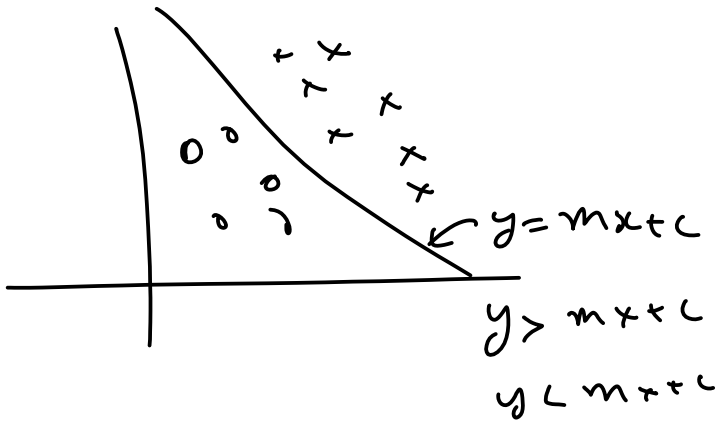
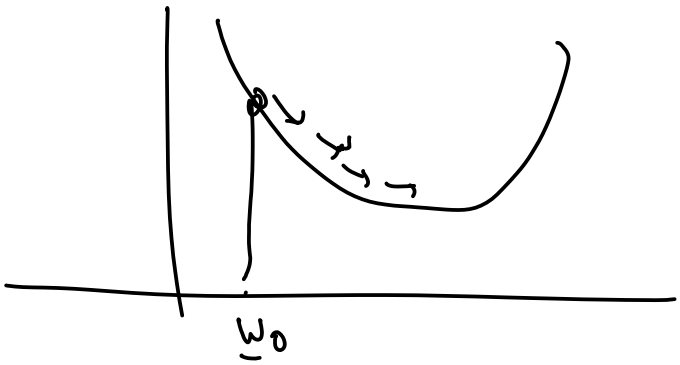
Perceptron and Gradient Descent



Gradient Descent



$$w_{t+1} = w_t - \alpha_t \nabla_{w_t} L(w_t; D_t)$$

Training dataset

Optimization perspective is not enough. We need a Probabilistic perspective.

(i.i.d)

The samples of Training Dataset must be independentally and identically distributed to the real world. (test Dataset)

$$D = \{(x_1, y_1), (x_2, y_2) \dots (x_i, y_i) \dots\}$$

$$P((x_i, y_i)) = P((x_j, y_j))$$ Identical

Def Two probability events being independent (A, B)

$$P(A \mid B) = P(A) \Leftrightarrow P(B \mid A) = P(B)$$

Def Conditional prob distribution

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A,B)}{\mathbb{P}(B)} \qquad \mathbb{P}(B) \neq 0$$

thm $A \perp B$ (↑ independent) if $\mathbb{P}(A,B) = \mathbb{P}(A)\cdot\mathbb{P}(B)$

Dataset is independentally distributed

$$(\underline{x}_i, y_i) \perp (\underline{x}_j, y_j)$$

$$\mathbb{P}(D) = \mathbb{P}((\underline{x}_1, y_1))\,\mathbb{P}((\underline{x}_2, y_1))\,\mathbb{P}((\underline{x}_3, y_3)) \ldots$$

$$= \prod_{i=1}^{n} \mathbb{P}((\underline{x}_i, y_i))$$

Maximum Likelihood estimation (ML)

Maximum-a-posteriori estimation (MAP)

BAYES THEOREM

Science	Machine Learning
Hypothesis	$\underline{w}$ Parameters
Evidence	$\underline{D}$ Dataset

$\underbrace{P(w|D)}_{\text{Posterior}} \stackrel{?}{=} \underbrace{P(D|w)}_{\text{Likelihood}}$

$$P(w|D) = \frac{P(w, D)}{P(D)} = \frac{P(D|w)\,P(w)}{P(D)}$$

Posterior — Likelihood — Prior — Evidence

$$P(w|D) = \frac{P(D|w)\,P(w)}{P(D)}$$

MAP estimates $\quad \arg\max_w P(w|D)$

$\arg\max_w P(D|w)\,P(w)$

ML estimate $\quad \arg\max_w P(D|w)$

How does this all relate to what we have been doing so far? Linear regression? Perceptron?

minimizing loss $\Leftrightarrow$ maximizing Probabilities

$l((x_i, y_i)) \qquad P((x_i, y_i))$

$= \|f_w(x_i) - y_i\|$

$\frac{1}{Z}\exp(-l_w((x_i, y_i))) = P((x_i, y_i)|w)$ $\quad Z = \int_x \int_y P(x_i, y_i)\,dx\,dy$

$$[0, \infty) \underset{\log(\,)}{\overset{\exp(-\,)}{\rightleftarrows}} (0, 1]$$

$$P(D|\underline{w}) = \prod_{i=1}^{n} P((\underline{x}_i, y_i)|\underline{w})$$

$$= \prod_{i=1}^{n} \frac{1}{Z} \exp\left(-\ell_{\underline{w}}((\underline{x}_i, y_i))\right)$$

$$= \frac{1}{Z} \exp\left(-\sum_{i=1}^{n} \ell_{\underline{w}}((\underline{x}_i, y_i))\right)$$

$$\underbrace{-\log\left(Z\, P(D|w)\right)}_{\text{Negative log likelihood (NLL)}} = \sum_{i=1}^{n} \ell_{\underline{w}}(\underline{x}_i, y_i)$$

Maximizing the likelihood is same as minimizing the negative log likelihood

---

Maximum a posteriori (MAP) brings in a Prior ($P(w)$).

$$\arg\max_{\underline{w}} P(\underline{w}|D) = \max_{\underline{w}} P(D|\underline{w})\, P(\underline{w})$$

$$\arg\min_{\underline{w}} -\log\left(P(\underline{w}|D)\right) = \arg\min_{\underline{w}} -\log\left(P(D|w)\right) - \log\left(P(\underline{w})\right)$$

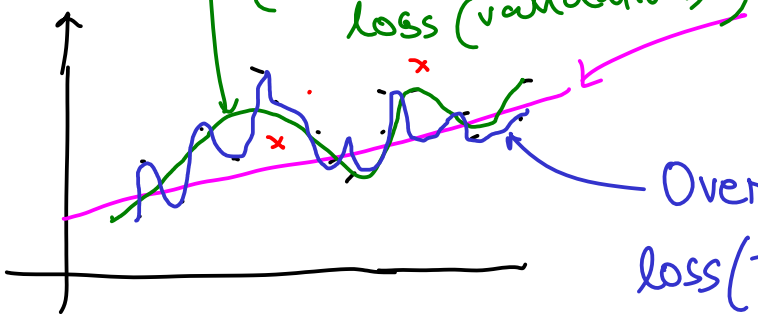
$$= \arg\min_{\underline{w}} \sum_{i=1}^{n} \ell_{\underline{w}}((\underline{x}_i, y_i)) + \underbrace{\lambda \|\underline{w}\|_2^2}_{\text{Prior knowledge}}$$

Occam's razor

Neural networks (Regularization)

⇕

(Overfitting)



minimize the loss on training set but not on test set

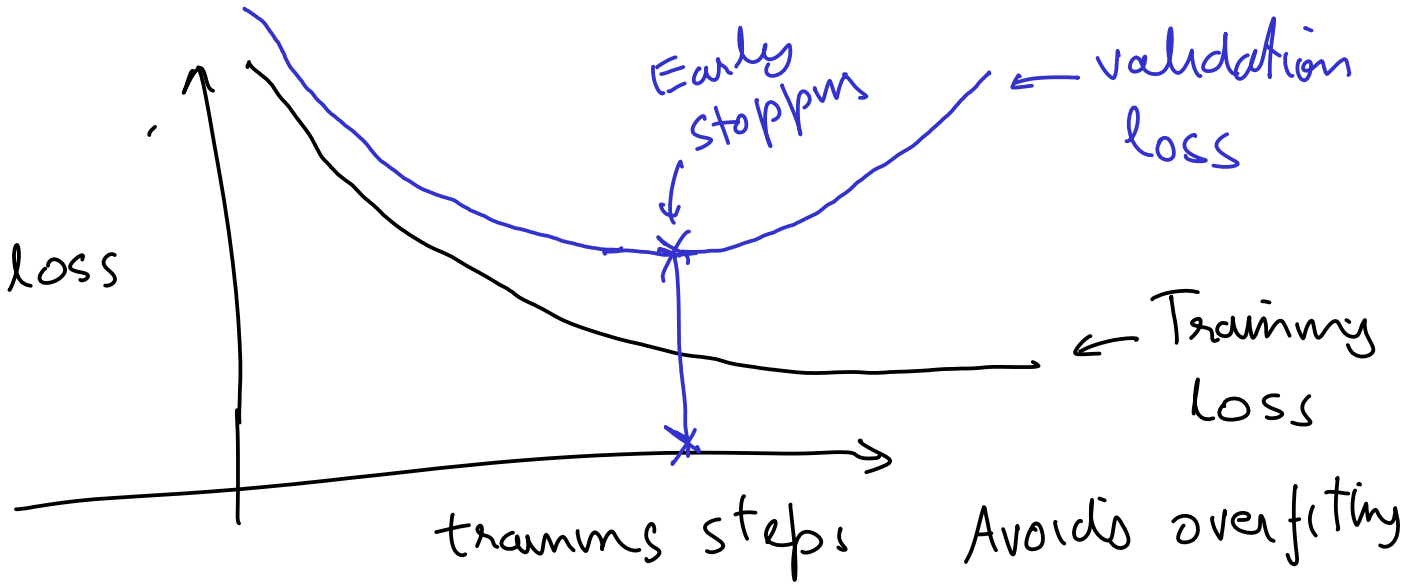
loss(Training) << loss(validation)

Early Stopping as a balance between Overfit and Underfit

D → Training 80 %
D → Validation 10 %

Gap between Training and Validation as a proxy for good fit

D → Test / Eval 10 %. Sacred ≡ proxy for the real world

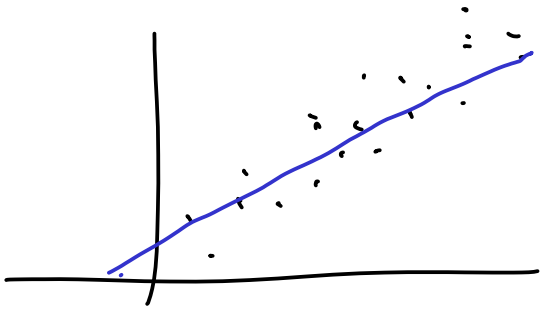


Avoids overfitting

Regularization (Linear regression)

$$\underset{\underline{w}}{\operatorname{argmin}} \; (\underline{y} - X\underline{w})^T (\underline{y} - X\underline{w}) + \lambda \|\underline{w}\|_2^2$$

$\underline{w} = \begin{bmatrix} m \\ c \end{bmatrix}$



$$\frac{\partial}{\partial \underline{w}} \lambda \|\underline{w}\|_2^2$$

$$= \frac{\partial}{\partial \underline{w}} \lambda \underline{w}^T \underline{w} = \frac{\partial}{\partial \underline{w}} \lambda \underline{w}^T \underline{I} \underline{w}$$

$$\frac{\partial}{\partial \underline{x}} \underline{x}^T A \underline{x} = \underline{x}^T (A + A^T)$$

$$= \lambda \underline{w}^T (\underline{I} + \underline{I}^T)$$

$$= 2\lambda \underline{w}^T$$

$$\underline{w}^* = (X^T X + \lambda \underline{I})^{-1} X^T \underline{y}$$